

Homework 1

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January 16, 2024

Problem 1. *Prove the Distributive Laws:*

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof. 1. $\forall x \in A \cap (B \cup C)$. By definition of a intersection, $x \in A$ and $x \in B \cup C$. By definition of union, $x \in B$ or $x \in C$. If $x \in B$, by definition of intersection, $x \in A \cap B$. If $x \in C$, by definition of intersection, $x \in A \cap C$. By definition of union, $x \in (A \cap B) \cup (A \cap C)$. Thus $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$. $\forall x \in (A \cap B) \cup (A \cap C)$, by definition of union, $x \in A \cap B$ or $x \in A \cap C$. By definition of intersection, $x \in A$ and $x \in B$ or $x \in C$. By definition of union, $x \in B \cup C$. Thus $x \in A$ and $x \in B \cup C$, by definition of intersection, $x \in A \cap (B \cup C)$. Thus $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$. Thus $(A \cap B) \cup (A \cap C) = A \cap (B \cup C)$.

2. $\forall x \in A \cup (B \cap C)$. By definition of union, $x \in A$ or $x \in B \cap C$. If $x \in A$, by definition of union, $x \in A \cup B$ and $x \in A \cup C$. If $x \in B \cap C$, then $x \in B$ and $x \in C$. By definition of union, $x \in A \cup B$ and $x \in A \cup C$. By definition of intersection, $x \in (A \cup B) \cap (A \cup C)$. Thus $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$. $\forall x \in (A \cup B) \cap (A \cup C)$. By definition of intersection, $x \in A \cup B$ and $x \in A \cup C$. By definition of union, since $x \in A \cup B$, $x \in A$ or $x \in B$. Similarly, $x \in A$ or $x \in C$. If $x \in A$, by definition of union, $x \in A \cup (B \cap C)$. If $x \notin A$, $x \in B$ and $x \in C$, then $x \in B \cap C$. Thus by definition of union, $x \in A \cup (B \cap C)$. Thus $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$. Thus $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

□

Problem 2. 2. Let $f(x) = x^2$ for $x \in \mathbb{R}$, and let $E = \{x \in \mathbb{R} : -1 \leq x \leq 0\}$ and $F = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

1. Show that $E \cap F = \{0\}$, and $f(E \cap F) = \{0\}$, while $f(E) = f(F) = \{y \in \mathbb{R} : 0 \leq y \leq 1\}$. Hence $f(E \cap F)$ is a proper subset of $f(E) \cap f(F)$.
2. Find $f^{-1}(E)$ and $f^{-1}(F)$, the inverse image of E , and the inverse image of F .

Solution. 1. *Proof.* $0 \in \mathbb{R}$ and $-1 \leq 0 \leq 0$ and $0 \leq 0 \leq 1$. So $0 \in E$ and $0 \in F$ so by definition of intersection, $E \cap F$. So $\{0\} \subseteq E \cap F$. $\forall x \in E \cap F$, by definition of intersection, $x \in E$ and $x \in F$. Suppose $x > 0$, then $0 < x \leq 1$. Thus $x \not\leq 0$, thus $x \notin E$, thus $x \notin E \cap F$. Similarly suppose, $x < 0$, then $-1 \leq x < 0$. Thus $x \not\geq 0$, thus $x \notin F$ and $x \notin E \cap F$. Thus $E \cap F = \emptyset$ or $E \cap F = \{0\}$. Thus $E \cap F \subseteq \{0\}$. Thus $E \cap F = \{0\}$.

By definition of image, $f(E \cap F) = f(\{0\}) = \{f(x) : x \in E \cap F\} = \{f(0)\} = \{0\}$. $f(E) = \{x^2 : -1 \leq x \leq 0\}$ and $f(F) = \{x^2 : 0 \leq x \leq 1\}$. Both sets represent non-negative numbers between 0 and 1. So $f(E) = f(F) = \{y : 0 \leq y \leq 1\}$. $f(F) = f(E) \implies f(E) \cap f(F) = f(E) = f(F)$. Since $0 \in f(F) \implies 0 \in f(E) \cap f(F) \implies f(E \cap F) \subset f(E) \cap f(F)$. ■

2. $\forall x \in E \setminus \{0\}, \exists a \in \mathbb{R} \ni f(a) = a^2 = x$, since f is strictly greater than or equal to 0 for all real numbers. For $x = 0$, only 0 is such a value such that $0^2 = 0$, because the square of any nonzero number is strictly positive. So $f^{-1}(E) = \{0\}$.

$\forall x \in F$, by definition $0 \leq x \leq 1$. For any positive real number $x, \exists \sqrt{x} \ni \sqrt{x}^2 = x$ and $(-\sqrt{x})^2 = x$. The square root of a number between 0 and 1 is also between 0 and 1. So $\exists \pm \sqrt{x} \in [-1, 1]$. So $f^{-1}(F) \subseteq [-1, 1]$. $\forall y \in [-1, 1], f(y) = y^2 \in F$, because the square of a real number between -1 and 1 is also between 0 and 1. So $y \in f^{-1}(F)$. So $[-1, 1] \subseteq f^{-1}(F) \implies f^{-1}(F) = [-1, 1]$. ■

Problem 3. Show that if $f : A \rightarrow B$, and G, H are subsets of B , then:

1. $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$.
2. $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$.

Solution. 1. $\forall x \in f^{-1}(G \cup H)$, by definition of inverse image, $f(x) \in G \cup H \implies f(x) \in G$ or $f(x) \in H$. If $f(x) \in G \implies x \in f^{-1}(G)$, by definition of union, $x \in f^{-1}(G) \cup f^{-1}(H)$. Similarly, if $f(x) \in H \implies x \in f^{-1}(H)$, by definition of union, $x \in f^{-1}(G) \cup f^{-1}(H)$. So $f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H)$.

$\forall x \in f^{-1}(G) \cup f^{-1}(H)$, by definition of union, $x \in f^{-1}(G)$ or $x \in f^{-1}(H)$. Thus $f(x) \in G$ or $f(x) \in H$ and by definition of union, $f(x) \in G \cup H$. By definition of inverse image. $x \in f^{-1}(G \cup H)$. Thus $f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H) \implies f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$

2. $\forall x \in f^{-1}(G \cap H)$, by definition of the inverse image, $f(x) \in G \cap H$. By definition of intersection, $f(x) \in G$ and $f(x) \in H$. By definition of inverse image, $x \in f^{-1}(G)$ and $x \in f^{-1}(H)$. By definition of intersection, $x \in f^{-1}(G) \cap f^{-1}(H)$. Thus $f^{-1}(G \cap H) \subseteq f^{-1}(G) \cap f^{-1}(H)$.

$\forall x \in f^{-1}(G) \cap f^{-1}(H)$, by definition of intersection, $x \in f^{-1}(G)$ and $x \in f^{-1}(H)$. By definition of inverse image, $f(x) \in G$ and $f(x) \in H$. By definition of intersection, $f(x) \in G \cap H$. By definition of inverse image, $x \in f^{-1}(G \cap H)$. Thus $f^{-1}(G) \cap f^{-1}(H) \subseteq f^{-1}(G \cap H)$. Thus $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$. ■

Problem 4. Prove that if $f : A \rightarrow B$ is bijective and $g : B \rightarrow C$ is bijective, then the composite $g \circ f$ is a bijective map of A onto C .

Proof. Injectivity: $\forall x, y \in A \ni (g \circ f)(x) = (g \circ f)(y) \implies g(f(x)) = g(f(y))$. By definition of bijective functions, g is injective. By definition of injective, since $g(f(x)) = g(f(y)) \implies f(x) = f(y)$. By definition of bijective functions, f is injective. By definition of injective functions, since $f(x) = f(y) \implies x = y$. Thus $g \circ f$ is injective.

Surjectivity: By definition of bijection, g and f are surjective. $\forall y \in C$, since g is surjective, $\exists b \in B \ni g(b) = y$. Since f is surjective, $\exists x \in A \ni f(x) = b$. Thus $g(f(x)) = g(b) = y$. Thus $(g \circ f)(x) = y$. Thus $g \circ f$ is a surjective function.

Thus by definition of bijective function, $g \circ f$ is a bijective map. □